HarvestMem: event-centric memory for vision–language–action harvesting robots under foliage occlusion

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Abstract

Selective harvesting robots must act on information they can no longer see: wind-blown foliage hides ripe fruits, and harvesting quotas depend on picks whose results lie outside the camera view. Vision–language–action (VLA) policies promise a single instruction-driven controller for such tasks, yet their memory is a fixed number of past frames sampled uniformly in time. We show that this time-indexed memory loses picks at a rate that grows with operating time: its coverage of completed picks decays inversely with elapsed time, and at a 16-frame budget it misses 43% of them. We propose HarvestMem, an event-centric memory framework that indexes memory by physical interaction events rather than by time. Its spatial memory, built by a SAM-based fruit distance module (SAM-FDM), anchors every fruit in 3D while it is visible. Its episodic memory, Picking-Event Keyframe Memory (PEKM), detects picks from the end-effector-to-fruit distance and reserves memory slots for their contact and detachment keyframes, without extra parameters, tokens or inference cost. On HarvestMem-Sim, a new benchmark with a 7-DoF arm, foliage occlusion and quota counting, PEKM retained 99.6% of completed picks versus 57.0% for uniform memory, and with 8 frames outperformed uniform memory with 16. Using only RGB-D images and proprioception, SAM-FDM localized fruits within 0.48 cm, recovered every leaf-hidden ripe fruit and reproduced the ground-truth picking events exactly, whereas anchors built after occlusion left no occluded episode correctly counted. [TBD: one sentence on closed-loop success]

Keywords: Selective harvesting, Vision–language–action model, Temporal memory, Keyframe selection, Foliage occlusion, Segment Anything Model, Imitation learning

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1. Introduction

Harvesting is among the most labour-intensive operations in fruit production, and the shrinking seasonal workforce has made robotic selective harvesting a long-standing goal of agricultural engineering (Bac et al., 2014; Kootstra et al., 2021; Zhou et al., 2022). Systems for sweet pepper (Arad et al., 2020), kiwifruit (Williams et al., 2019) and apple (Silwal et al., 2017) are pipelines of detection, localization, planning and grasping modules engineered for one crop, one decision rule and one scene layout. Vision–language–action (VLA) policies offer a different paradigm: a single network pretrained on web-scale vision–language data and large robot datasets maps images, proprioception and a natural-language instruction directly to actions (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024; Physical Intelligence et al., 2025), so that *which* fruits to pick, *how many* and *where* to put them become an instruction rather than an engineering effort.

Harvesting is a memory problem. A harvesting robot rarely has in view the information it needs for its next decision. Wind or the manipulator moves leaves over ripe fruits that were visible moments earlier (Hemming et al., 2014); grading, sampling and thinning require a given number of fruits; and the collection bin is usually outside the camera's view, so earlier picks cannot be re-observed. Forgetting has a direct agronomic cost: a forgotten ripe fruit is lost yield, a forgotten pick violates the quota. In the benchmark introduced here, a leaf-hidden ripe fruit is invisible in a median 78% of the frames in which it still hangs on the tree, so the decision to pick it must be taken largely from memory.

Time-indexed memory forgets more the longer the robot works. Memory-augmented VLA policies keep a bounded number of past frames, because every frame adds visual tokens and attention cost grows quadratically with their number (Vaswani et al., 2017), and they fill this budget by sampling the history uniformly in time. This design has a structural flaw for manipulation. The frames that define a pick, the gripper closing on a fruit and the fruit detaching, span only about a dozen frames, and fewer than 5% of uniformly sampled frames show the gripper within 3 cm of a fruit to be picked. With B frames spread over t elapsed frames, a pick is retained with probability roughly $(B-1)|S|/t$, where $|S|$ is the length of its key window, so coverage

decays inversely with operating time (Section 4.4). At a 16-frame budget, uniform memory misses 43% of completed picks, recovers the pick count at only 43% of decisions and, after 600 frames, loses even the *most recent* pick at 78% of decisions. These losses matter: on a general manipulation-memory benchmark, removing the memory of the same base VLA reduces success from 66.7% to 1.7%, and halving its budget reduces success from 64.4% to 26.7% (Section 6.6).

Why the obvious remedies fall short. Enlarging the memory treats the symptom: a uniform budget must grow linearly with operating time, at a quadratic attention cost that embedded field computers cannot afford. *Better per-frame perception* cannot see what is hidden and mistakes a carried fruit for a new target; it recovers the correct pick count in only 62% of our episodes. *Learned memory selection or token compression* adds parameters and training signals, may behave differently in training and deployment, and is hard to audit. What is missing is a *principle* that tells a harvesting robot what is worth remembering.

Key insight: index memory by events, not by time. In agricultural manipulation, task-relevant information is concentrated at a few physical interaction events. Every pick consists of an approach, a contact and a detachment; these moments change the state of the task, and they are physically observable, because the distance between the tool and the nearest fruit still to be picked drops below the fruit radius at contact and jumps at detachment. This mirrors human memory, which segments experience into discrete events and preferentially encodes their boundaries (Zacks et al., 2007), and whose episodic component binds *what* happened to *where* and *when* (Tulving, 1972; Clayton and Dickinson, 1998).

HarvestMem. We introduce *HarvestMem*, which equips a VLA harvesting policy with two complementary memories coupled through one physical signal (Fig. 1). Its **spatial memory** answers *where* the fruits are: a SAM-based fruit distance module (SAM-FDM) segments fruits with the Segment Anything Model (Kirillov et al., 2023) while they are visible and anchors each one in 3D, persisting under foliage. Its **episodic memory** answers *what* happened and *when*: Picking-Event Keyframe Memory (PEKM) detects picks causally from the tool-to-fruit distance and reserves slots in the VLA's fixed memory for their contact and detachment keyframes. HarvestMem changes

what the policy remembers, not *how much* or *how*: PEKM adds no parameters, tokens or inference cost, contains the base model's uniform memory as the special case $k = 0$, and is applied identically in training and deployment. Phase-aware recovery demonstrations (DART-H) complement the memory by addressing the compounding errors of closed-loop imitation (Ross et al., 2011).

Contributions.

1. **Problem formulation and benchmark.** We formulate selective harvesting with a VLA policy as a fixed-budget memory-allocation problem and release HarvestMem-Sim, a procedural ManiSkill3 orchard with a commercial 7-DoF arm and two tasks that cannot be solved from the current view: picking leaf-hidden ripe fruits (FruitOccludedRipe) and meeting a quota with an out-of-view basket (FruitCount).
2. **Event-centric memory with a coverage analysis.** We show that time-indexed memory covers interaction events with a probability that decays inversely with operating time (predicted to within 0.012), and propose PEKM, which guarantees coverage of the most recent picks at any episode length and retains 99.6% of completed picks against 57.0% for uniform memory, at zero additional cost.
3. **Spatial–episodic coupling through promptable segmentation.** SAM-FDM turns RGB-D images and proprioception into 3D fruit anchors and picking evidence, reproduces the ground-truth events exactly, and shows that spatial memory acquired before occlusion is a *precondition* for episodic memory; a robustness analysis gives perception an explicit accuracy specification.
4. **Phase-aware recovery demonstrations (DART-H)**, with correlated joint noise injected only in free-motion phases and clean planner actions as labels.
5. **A rigorous evaluation protocol**, sealed before the formal runs, with paired bootstrap, Holm correction, equivalence margins, a per-decision audit of the frames fed to the model, and counterfactual controls that separate event information from the mere change of frames.

2. Related work

Harvesting robots, fruit perception and occlusion. Reviews identify fruit detection under occlusion, localization accuracy and cycle time as the main

[TBD: Fig. 1 (to be drawn). Left: the memory needs of harvesting (ripe fruit hidden by wind-blown leaves; quota with an out-of-view basket); uniform memory misses the short pick events. Right: front/wrist RGB-D → SAM-FDM (colour prompts, SAM masks, 3D anchors, vacancy + grasp evidence; *where*) → d_t → event detector → PEKM keyframes (*what*/*when*) → VLA memory expert → action chunk; the same selection function computes the memory of every training sample.]

Figure 1: HarvestMem indexes memory by physical events instead of time. Fruits are anchored in 3D while visible (spatial memory), and the contact and detachment keyframes of picks are reserved in the VLA's fixed memory budget (episodic memory). The distance d_t is the only interface between perception and the policy; it selects frame indices and never enters the network.

bottlenecks of harvesting robots (Bac et al., 2014; Kootstra et al., 2021; Zhou et al., 2022), whose field systems follow a detect–localize–plan–grasp pipeline. Deep networks such as Mask R-CNN (He et al., 2017) and YOLO (Redmon et al., 2016) have replaced hand-crafted colour and shape features (Sa et al., 2016; Kamilaris and Prenafeta-Boldú, 2018; Koirala et al., 2019), often combined with depth or multi-view geometry for 3D localization (Gené-Mola et al., 2020; Kang and Chen, 2020). Promptable segmentation models, SAM and SAM 2 (Kirillov et al., 2023; Ravi et al., 2024), give zero-shot masks from point or box prompts and are increasingly used in agriculture [TBD: cite verified agricultural SAM works]. Occlusion by leaves is a primary cause of missed fruits (Hemming et al., 2014); multi-view cameras, next-best-view planning and foliage manipulation [TBD: cite verified works] address what is not visible in the *current* view, but not fruits that were seen earlier and are hidden now, which calls for memory rather than a better viewpoint. SAM-FDM uses SAM in promptable mode, but its output is a distance signal that organizes the memory of a learned policy, not a detection map for a planner.

VLA policies and imitation learning. VLA models extend vision–language models with action outputs: RT-2 (Brohan et al., 2023), OpenVLA (Kim et al., 2024), Octo (Octo Model Team et al., 2024), and the flow-matching

Table 1: Memory designs for a VLA policy under a fixed budget. ✓: yes; ×: no; ○: only by chance or implicitly.

Memory design	Distant past	Pick events kept	Fixed cost	No extra params	Train = deploy	Physically grounded
No memory (current frame)	×	×	✓	✓	✓	×
Recent window	×	○	✓	✓	✓	×
Uniform sampling (base model)	✓	○	✓	✓	✓	×
Learned memory bank / compression	✓	○	✓	×	✓	×
PEKM (HarvestMem)	✓	✓	✓	✓	✓	✓

policies π_0 and $\pi_{0.5}$ (Black et al., 2024; Physical Intelligence et al., 2025), built on PaliGemma (Beyer et al., 2024), SigLIP (Zhai et al., 2023) and flow matching (Lipman et al., 2023). Policies trained from demonstrations, including Diffusion Policy (Chi et al., 2023) and ACT (Zhao et al., 2023), suffer from covariate shift (Ross et al., 2011), remedied by interactive corrections (DAgger) or by noise injection during demonstration (DART, Laskey et al., 2017), which DART-H follows because it needs no expert in the loop. Applications of VLA policies to agricultural manipulation are still rare [TBD: cite verified early agricultural VLA works, if any].

Memory and keyframes in robot policies. Long-horizon manipulation benchmarks show that policies without history fail on tasks that require recalling past observations [TBD: cite RoboMME (verify)], and memory-augmented VLAs keep a bounded history of visual tokens [TBD: cite MME-VLA (verify)]. Keyframes are used in manipulation as action targets (Shridhar et al., 2022) and as waypoints that compress demonstrations (Shi et al., 2023); in video understanding, memory banks compress long videos for multimodal models (He et al., 2024). To our knowledge, no prior work selects the memory frames of a VLA policy from task-defined physical events, nor studies memory selection for agricultural manipulation.

Positioning. Table 1 contrasts memory designs for a VLA policy under a fixed budget. Time-indexed designs keep the recent past or the whole past at a resolution that falls with elapsed time, so short interaction events are kept only by chance; learned memory banks and token compression discover what matters only implicitly, with additional parameters, and are hard to audit. HarvestMem differs in three ways: (i) the unit of memory is a *physical interaction event* defined by task geometry; (ii) selection is a parameter-free deterministic function, identical in training and deployment, that contains uniform memory as a special case; and (iii) episodic memory of events is coupled to an explicit *spatial* memory of objects acquired before occlusion, a coupling that Section 6.4 shows to be necessary.

3. HarvestMem-Sim: a memory-dependent harvesting benchmark

Scene, robot and sensing. HarvestMem-Sim is built on ManiSkill3 (Tao et al., 2025) (environment `FruitPick-v1`). Each episode samples a tree with a trunk, branches and leaf clusters, and 4–6 spherical fruits of radius 2.5–2.9 cm at heights of 0.44–0.64 m, distances of 0.38–0.50 m from the robot base and within $\pm 28^\circ$ of azimuth. The layout sampler guarantees an inverse-kinematics solution for a horizontal approach, a branch-free approach corridor and no mutual occlusion of fruits in the front camera. A fruit is attached to its stem until it has been held between both fingers for two consecutive control steps; it then becomes a dynamic rigid body. The robot is a RealMan RM75-B 7-DoF arm with an EG2 parallel gripper under joint-position control (a 6-DoF RM65-B is also supported). Observations are a front and a wrist RGB-D camera (256×256 pixels; front field of view 55°) and an 8-dimensional state (seven joint angles and one finger position); actions are seven absolute joint targets and a gripper command.

Tasks. Figure 2 shows both tasks. In **FruitOccludedRipe**, red (ripe) and green (unripe) fruits hang together. During a 30-step observation phase, a gust between steps 8 and 20 blows two leaf clusters in front of two fruits, at least one of them ripe, so "hidden" does not imply "ripe". The instruction is to put all ripe fruits into the basket. Leaves are pushed aside when the gripper comes within 15 cm, so the final approach is visible, but *finding* a hidden ripe fruit requires remembering it. Success requires every ripe fruit and no unripe fruit in the basket; a dropped fruit is a failure. In **FruitCount**, all M fruits are ripe and the instruction asks for exactly $N < M$ of them ($N = 2$–5). The basket is outside the front camera view, so the policy can only know how many it has picked by remembering its own picks. Success requires exactly N fruits in the basket; over-picking or dropping is a failure.

Demonstrations. A scripted demonstrator plans with mplib (Hao Su Lab, 2024) through eight phases (observe, reach, approach, grasp, pull, carry, release, settle); failed attempts are discarded, and every frame stores the ground-truth fruit poses and attachment, visible pixels per camera, phase labels and TCP pose. Recovery demonstrations (DART-H, Section 4.7) use the same demonstrator with execution noise. Table 2 summarizes the 640 demonstrations. After the gust, the targeted fruits are fully invisible in the front camera; over all frames in which an occluded fruit is still on the tree, it is invisible in 72% of frames on average (median 78%), and two-thirds of

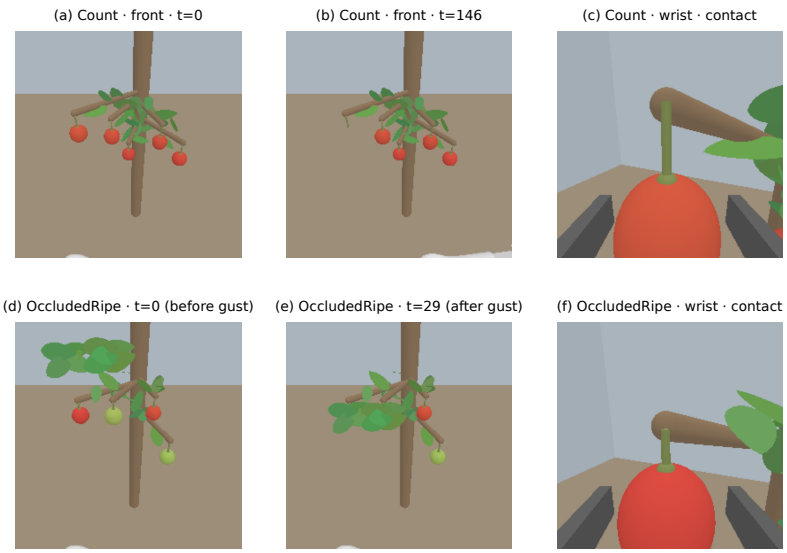


Figure 2: HarvestMem-Sim tasks. (a–c) FruitCount: front view at the start and after one pick, and the wrist view at contact. (d–f) FruitOccludedRipe: the front view before and after the wind gust (two leaf clusters now hide fruits) and the wrist view at contact.

Table 2: Benchmark statistics (steps: mean ± SD). Acceptance: fraction of demonstrator attempts that succeeded. Training features: 88,607 samples (clean) and 173,791 (clean + DART-H).

Set (episodes)	Task	n	Steps	Range	Fruits	Picks	Acceptance
Clean train (0–99)	FruitCount	100	474 ± 124	320–757	4.73	2.85	0.96
	FruitOccludedRipe	100	412 ± 85	295–710	4.69	2.54	0.98
DART-H train (100–199)	FruitCount	100	455 ± 133	314–876	4.62	2.70	0.57
	FruitOccludedRipe	100	397 ± 89	297–672	4.78	2.46	0.66
Development (200–219)	FruitCount	20	487 ± 162	321–766	5.05	3.10	1.00
	FruitOccludedRipe	20	369 ± 71	300–471	4.60	2.35	0.95
Validation (0–49)	FruitCount	50	459 ± 142	314–904	4.58	2.82	0.98
	FruitOccludedRipe	50	396 ± 87	300–630	4.72	2.50	0.96
Validation (50–99)	FruitCount	50[a]	499 ± 169	322–1037	4.67	2.98	0.91
	FruitOccludedRipe	50	392 ± 73	304–534	4.54	2.52	0.98

[a]Statistics over 49 episodes; the ground-truth file of one episode was regenerated after these statistics were computed.

the leaf clusters hide a ripe fruit (Table S1 in the Supplementary material, with the phase structure).

4. HarvestMem: event-centric memory for harvesting

4.1. Design principles

HarvestMem rests on four design principles. **P1, index memory by events, not by time:** memory slots go to the contact and detachment of each pick, which time-indexed memory provably loses as operation lengthens (Section 4.4). **P2, ground events in physics:** events are defined by the tool-to-fruit distance, which is measurable on a real robot, interpretable, and transferable across crops through one parameter commensurate with fruit size. **P3, change what is remembered, not how much or how:** budget, tokens, architecture and inference cost of the base VLA are unchanged, and the same selection is used in training and deployment. **P4, remember where before remembering what:** an event on a hidden fruit can be recognized only if the fruit was anchored while visible, so spatial memory is built first and drives episodic memory. In the resulting system (Fig. 1), SAM-FDM (Section 4.6) builds the spatial memory and the distance d_t, a causal detector (Section 4.3) finds picking events, and PEKM (Section 4.5) chooses which B past frames the VLA sees.

4.2. Problem formulation and base policy

At decision time t, the policy π_θ receives the current front and wrist images o_t, the proprioceptive state s_t, the instruction l and a memory $\mathcal{M}_t$ of past frames, and outputs an action chunk $A_t = (a_t, \ldots, a_{t+H-1}) = \pi_\theta(o_t, s_t, l, \mathcal{M}_t)$ with $H = 20$, of which the first 16 actions are executed before the next decision. The memory is the output of a selection function

$$\mathcal{M}_t = S\big(\{0, \ldots, t\};\, d_{0:t}\big), \qquad |\mathcal{M}_t| = \min(B, t+1), \tag{1}$$

that is *causal* and *budget-preserving* (exactly B frames, as with uniform sampling), so any difference in behaviour is due to memory content rather than memory size.

The base policy is a memory-augmented VLA (MME-VLA, [TBD: cite base model]) in the $\pi_{0.5}$ family (Physical Intelligence et al., 2025): a PaliGemma backbone (Gemma-2B, Beyer et al., 2024), a 300M-parameter action expert that generates action chunks by flow matching (Lipman et al., 2023), and a 150M-parameter memory expert that attends over the $B = 16$ selected frames, each encoded by the frozen SigLIP encoder (Zhai et al., 2023) into 16 tokens. We initialize from a checkpoint trained on a general manipulation-memory benchmark ([TBD: cite RoboMME]) and fine-tune all non-vision parameters on HarvestMem-Sim.

4.3. Event signal and causal event detection

The event signal is

$$d_t = \min_{j \in \mathcal{P}_t} \| p_t^{\mathrm{TCP}} - c_j \|_2, \tag{2}$$

where p_t^{TCP} is the TCP position from forward kinematics, c_j the centre of fruit j, and $\mathcal{P}_t$ the set of fruits that are *pickable and still attached* (all fruits in FruitCount, ripe fruits in FruitOccludedRipe). If $\mathcal{P}_t$ is empty, $d_t = 10$ m; a frame without a measurement is unknown. The signal falls below the fruit radius when the gripper encloses a fruit and jumps at detachment, because a detached fruit leaves $\mathcal{P}_t$. In training, d_t comes from the simulator state; at deployment, from SAM-FDM with the same definition.

A frame is *near* if $d_t \le r$ and *far* otherwise; an unknown frame inherits the last known state, so a dropped measurement neither starts nor ends an event. An *event* is a maximal run of near frames $[t_s, t_e]$ and contributes two keyframes:

$$t^{\mathrm{c}} = \min \Big\{ f \in [t_s, \min(t_e, t)] : d_f \le \min_{t_s \le g \le \min(t_e, t)} d_g + \mathrm{tol} \Big\}, \tag{3}$$

$$t^{\mathrm{r}} = t_e + 1 \quad (\text{only if } t_e < t). \tag{4}$$

The contact keyframe is the first frame within tol of the minimum distance of the run observed so far (the gripper settling around the fruit), and the release keyframe is the first far frame after the run (the detachment). We use $r = 3$ cm, commensurate with the fruit radius and below half the inter-fruit spacing, and tol $= 5$ mm, both chosen on the training set. For noisy deployment signals, a *hysteresis* variant enters an event at $d_t \le r$, leaves it only at $d_t > 1.5r$ and discards runs shorter than 3 frames, which suppresses chatter around the threshold without the delay of a median filter.

4.4. Why time-indexed memory loses events: a coverage analysis

Let the key set S_e of a completed pick e be the frames within $\pm W$ of its contact and release keyframes, $|S_e| \le 2(2W + 1)$; the pick is *covered* at decision time t if the memory holds a frame of S_e. Uniform memory keeps one frame every $\Delta_t = t/(B - 1)$ frames. If the phase of this grid is independent of the timing of the pick, the expected number of memory frames in S_e is $|S_e|/\Delta_t$, and when Δ_t exceeds each keyframe window, at most one memory frame falls in each window, so that

$$c_U(t) \approx \min\Big(1,\ \frac{(B-1)\,|S_e|}{t}\Big). \tag{5}$$

Algorithm 1 PEKM selection at decision time t

Require: distances $d_{0:t}$, budget B, keyframe cap $k < B$, radius r, tolerance tol

if $t+1 \le B$ **then return** $\{0, \dots, t\}$
end if
$\mathcal{E} \leftarrow \textsc{DetectEvents}(d_{0:t}, r, \text{tol})$ ▷ Eqs. (3)–(4)
$\mathcal{K} \leftarrow \emptyset$
for $e \in \mathcal{E}$ from newest to oldest **do**
  for $f \in (t_e^{\mathrm{c}}, t_e^{\mathrm{r}})$ if defined **do**
    if $|\mathcal{K}| < k$ and $f \notin \mathcal{K}$ **then** $\mathcal{K} \leftarrow \mathcal{K} \cup \{f\}$
    end if
  end for
end for
$R \leftarrow \{0, \dots, t\} \setminus \mathcal{K}$ ▷ sorted
$U \leftarrow \{R[\lfloor i\,(|R|-1)/(B-|\mathcal{K}|-1)\rfloor] : i = 0, \dots, B-|\mathcal{K}|-1\}$ ▷ evenly spaced
return sort$(\mathcal{K} \cup U)$

Two consequences are central to harvesting. First, **forgetting grows with operating time**: for a fixed budget, coverage decays as $1/t$, and the most recent pick enjoys no protection. Second, **budgets must grow linearly with time**: to keep coverage c, uniform memory needs $B \ge 1 + c\,t/|S_e|$; with $W = 3$ ($|S_e| = 14$) and $t = 600$, full coverage would require about 44 frames, nearly three times the base model's budget, at a quadratically growing attention cost. Event indexing removes the dependence on t: with keyframe cap k, PEKM covers the $\lfloor k/2 \rfloor$ most recent picks with probability one at every decision time, using a constant two slots per remembered pick.

4.5. Picking-Event Keyframe Memory

Algorithm 1 gives the selection rule: keyframes are taken from the most recent event backwards, contact before release, until at most k are reserved, and the remaining $B - |\mathcal{K}|$ slots are filled at evenly spaced positions among the non-key frames, including the first and the last.

Properties. PEKM is a drop-in memory for a pretrained VLA. It is *budget-preserving* ($|\mathcal{M}_t| = B$), so compute and latency are unchanged. It *strictly nests the baseline*: with $k = 0$ it is exactly the uniform sampling of the base model. It is *causal*: an ongoing event contributes a provisional contact

frame. It is *train–deploy consistent*: every training memory is computed with the same function and signal definition as at deployment. It provides *signal isolation*: d_t selects frame indices but never enters the network, so a perception error can change *which* frames are seen but cannot inject a wrong value into the policy. With $k = 4$ the memory holds the keyframes of the two most recent picks, which show which fruit is gone and that a basket drop has occurred; $k = 8$ covers four picks, enough for every FruitOccludedRipe episode and nearly every FruitCount episode.

Soft variant (PEKM-S). As an alternative to hard reservation, PEKM-S samples the B frames from the density $\rho_f = (1-\alpha)/N + \alpha\, w_f / \sum_g w_g$ over the N history frames, with proximity weights $w_f = \exp(-d_f/\tau_d)$ and $\tau_d = 5$ cm, so that $\alpha \in [0, 1]$ interpolates between uniform ($\alpha = 0$) and pure proximity memory ($\alpha = 1$); sampling details are given in Supplementary Section S4.

4.6. SAM-based fruit distance module

On a real robot, d_t must come from perception. SAM-FDM produces it from the front RGB-D camera, forward kinematics and the gripper state in four steps.

(i) Colour-prompted segmentation. Ripe (red) and unripe (green) candidate regions are found by HSV thresholds and connected components (calibrated on training renders; pixel F1 0.986 and 0.976). Each component prompts SAM (Kirillov et al., 2023) with its bounding box padded by 2 pixels and its centroid as a positive point, keeping the mask with the highest predicted IoU. Masks much larger than their prompt region or of mixed colour (purity < 0.3) are rejected, duplicates are removed by mask non-maximum suppression, and ripeness is the dominant colour class inside the mask.

(ii) Fruit anchors (spatial memory). At the first frame, before any occlusion, every mask is back-projected with the depth image and camera parameters, and the fruit centre is the median visible surface point pushed back along the viewing ray by the fruit radius,

$$\hat{c}_j = \tilde{p}_j + \hat{\rho}_j \frac{\tilde{p}_j - o_{\text{cam}}}{\|\tilde{p}_j - o_{\text{cam}}\|}, \qquad \hat{\rho}_j = \frac{z_j}{f_x}\sqrt{\frac{|m_j|}{\pi}}, \tag{6}$$

where $\tilde{p}_j$ is the median back-projected point of mask m_j, z_j its median depth, f_x the focal length and o_{cam} the camera centre; anchors closer than 3 cm are

merged. Each anchor stores its centre and ripeness and persists whether or not the fruit remains visible, which is how the module remembers a ripe fruit that wind later hides. SAM therefore runs only once per episode.

(iii) Picked-fruit detection from two kinds of evidence. An anchor is marked *picked* by either of two causal cues. *Depth vacancy* (visual): the median depth around the anchor's projected surface point lies more than 5 cm behind the expected surface for 3 consecutive frames (more than 2 cm in front means occluded, and the anchor is left unchanged). *Grasp holding* (proprioceptive): the gripper is commanded closed but the fingers stop between the empty-closed and open positions for 2 frames while the TCP is within r of an unpicked anchor. Vacancy needs no contact sensing but is blind while the arm occludes the anchor during carrying; grasp holding fires exactly at detachment.

(iv) Distance. $\hat{d}_t$ is the distance from the TCP to the nearest unpicked pickable anchor, exactly as in Eq. (2), and feeds the hysteresis detector and PEKM. Through this scalar interface, SAM can be replaced by any detector (e.g. YOLO-seg or a crop-specific network) without retraining the policy.

4.7. Phase-aware recovery demonstrations (DART-H)

Imitation policies drift away from the demonstrated states in closed loop (Ross et al., 2011). Following DART (Laskey et al., 2017), we inject noise while the demonstrator executes but record its clean actions, replanned from the perturbed state, as labels that show how to recover. Two choices adapt DART to harvesting. First, white joint noise is filtered out by the position controller, so we use *temporally correlated* Ornstein–Uhlenbeck noise on the seven joint targets,

$$\epsilon_{t+1} = \epsilon_t - \frac{\epsilon_t}{\tau} + \sigma\sqrt{\frac{2}{\tau}}\,\xi_t, \qquad \xi_t \sim \mathcal{N}(0, I), \tag{7}$$

with $\sigma = 0.04$ rad and $\tau = 10$ steps, which produces median TCP displacements of 2.50 cm (FruitCount) and 2.54 cm (FruitOccludedRipe), the scale of the drift observed in closed loop. Second, the noise is *phase-gated*: scaled by 1 in reach, pull and carry, 0.5 in approach, and 0 in observe, grasp, release and settle, so grasping and releasing stay clean. Failed perturbed attempts are discarded, and the training set combines 200 clean and 200 DART-H episodes.

All memory conditions are fine-tuned with identical settings and differ only in the keyframe cap k used to compute the memory of each training sample: 12,000 steps, global batch size 64, AdamW, peak learning rate 2.5×10^{-5} with warm-up and cosine decay, EMA decay 0.999 and a frozen vision encoder (details in Supplementary Section S7).

5. Experimental design

5.1. Research questions and experiment map

The experiments address eight research questions (Table 3). E1–E6 and E9–E11 analyse the components offline and in perception; E7 and E8 evaluate closed-loop harvesting under the pre-registered protocol of Section 5.5.

5.2. Data and splits

Training, validation and test seeds come from disjoint ranges, so every evaluated layout is unseen in training, and all method parameters were fixed on training data or by physical reasoning, never on validation data (Supplementary Table S9). Because validation episodes 0–49 were used in earlier development, the formal closed-loop evaluation (E7, E8) uses episodes 50–99, untouched by any evaluation or tuning; development episodes 200–219 serve only as a go/no-go gate. The E5 perception set replays validation episodes 0–49 and renders front and wrist RGB-D images with ground-truth instance masks every 10th frame (8644 images); thresholds and prompts were calibrated on 20 rendered training episodes. Offline analyses use 199 of the 200 validation episodes (one unreadable ground-truth file was later regenerated; the coverage-by-time analysis uses all 200).

5.3. Compared methods

Memory selection (E4, E7, E8). All memories have exactly B frames: **Uniform** (U, K0), the base model's even sampling, identical to PEKM with $k = 0$; **Recent**, the last B frames; **Random**, frames 0 and t plus $B - 2$ random frames; **PEKM** (Kk), $k \in \{2, 4, 8, 12\}$ offline and $k \in \{4, 8\}$ trained for closed loop; and **PEKM-S**, $\alpha \in \{0.25, 0.5, 1\}$.

Table 3: Research questions and experiment map. C1–C5: contributions in Section 1.

Exp. (RQ)	Question	Data	Main metrics	Contr.
E1 (RQ1)	Do the tasks require memory; how strong is the occlusion?	640 demonstrations	Steps, picks, occlusion	C1
E2 (RQ2)	Does the tool-to-fruit distance locate picks; how sensitive is it to r, tol?	Train 200 (selection), val 199, DART-H 200	Event P/R/F1, count exact, keyframe timing	C2
E3 (RQ3)	How robust are events and memory to noise, dropout and bias?	Val 199, 5 seeds per level	Event F1, count exact, keyframe recall	C2, 3
E4 (RQ4)	What do uniform memory and PEKM retain; does the coverage model hold?	Val 199–200, 5318 decisions	Event coverage, count recoverability, recency, gap	C2
E5 (RQ5)	Can SAM segment fruits and ripeness under occlusion and lighting shifts?	8644 rendered images with GT masks	Instance F1, IoU, recall by visibility, latency	C3
E6 (RQ6)	Does RGB-D $\rightarrow$ SAM-FDM $\rightarrow$ events reproduce the GT memory?	Val 199 RGB-D + proprioception	Anchor/distance error, event F1, memory agreement	C3
E7 (RQ7)	Does event memory improve closed-loop harvesting?	Val 50–99 $\times$ 3 seeds	Success, completion, wrong/over-pick, drop	C2, 5
E8 (RQ7)	Is the gain due to event information?	As E7	Paired differences to EV	C2, 3, 5
E9 (RQ8)	Where do policies fail in closed loop; does DART-H help?	10 episodes per set; E7	TCP error by horizon; C0 vs K0	C4
E10 (–)	Generalization and supporting evidence	Unseen layouts; memory benchmark	Success by quota; ablations	C2
E11 (RQ8)	What does the approach cost at run time?	CPU / L20 GPU	Selection and SAM latency	C2, 3

Counterfactual controls (E8). A change of memory content can affect a policy simply because the frames differ. Each event memory (EV) is therefore compared with three controls that keep the budget, the first frame and the recent window: **EVU**, the same sampler with a uniform density; **EVR**, uniform memory with exactly the frames that EV dropped replaced by random ones; and **EVA**, the anti-event selection $(1 - w_f)$. Only EV$-$EVU and EV$-$EVR differences favouring EV, with EVA no better than U, count as an effect of event information.

Detection, segmentation and perception chain. Event detectors are plain thresholding, a causal 5-frame median filter (median5) and hysteresis. Segmentation (E5) compares **HSV** colour thresholds, **SAM-AUTO** (zero-shot automatic masks filtered by colour and compactness), **SAM-FDM** with ViT-B/L/H, and **SAM-GTBOX** (ground-truth box prompts, an upper bound). The perception chain (E6) compares **GT**, **SAM-FDM**, SAM-FDM with consumer-grade **depth noise**, **depth-only** (no grasp evidence), **post-gust anchors** built at frame 29 (no spatial memory before occlusion) and **per-frame** detection without anchors. For DART-H (E9), C0 (clean data only) is compared with K0 (clean plus DART-H).

5.4. Metrics

The primary closed-loop metric is the **success rate** (Section 3); process metrics are harvest completion, wrong-pick, over-pick and drop rates and steps, and engineering errors are reported separately. Detected events are matched one-to-one to ground-truth detachments within ± 3 frames of the event; we report precision, recall, F1 and the **count exact rate** (episodes whose number of detected events equals the number of picks). At every decision point, a completed pick is **covered** if the memory holds a frame within ± 3 frames of its contact or release keyframe; **event coverage** is the fraction of covered picks, **count recoverability** the fraction of decisions at which every completed pick is covered, and **recency coverage** the coverage of the last three picks. Segmentation is scored by instance F1 at IoU $\geq$ 0.5, mean IoU, ripeness accuracy and recall by visible fraction; anchors and distances by their errors against ground truth. Full definitions are given in Supplementary Section S7.

5.5. Closed-loop evaluation protocol and statistics

The closed-loop protocol was sealed (SHA-256 digest recorded) before the formal runs, and the first completed run is the result, without re-runs or selection. Validation episodes 50–99 of each task are evaluated with three policy sampling seeds, giving 150 trials per task and condition (episode limit 1300 steps). The conditions are K0, K4 and K8 (trained on clean plus DART-H data) and C0 (uniform, clean data only). For each episode and seed, all conditions run in one block with the same initial state, in randomized order. A run is valid only if engineering errors are at most 2% of trials, every block is complete, and a per-decision audit confirms that the frame indices fed to the model equal those recomputed offline by the declared selection

rule. Beforehand, K0 must pass a development gate (success above 5%), so a broken pipeline cannot consume the validation episodes.

Differences are paired differences in success rate, with confidence intervals from a cluster bootstrap over episodes (20,000 resamples). The primary contrasts K4−K0 and K8−K0 per task are Holm-corrected (Holm, 1979), and a difference is called *equivalent* if its 90% confidence interval lies within ±12 percentage points (two one-sided tests, Schuirmann, 1987). Secondary contrasts (K0−C0; counterfactual controls) are reported with unadjusted intervals. Offline experiments are deterministic except for noise and random baselines (5 seeds).

6. Results

The benchmark requires memory (RQ1). Episodes last 300 to 1000 control steps with 2–5 picks (Table 2), while a pick, from contact to detachment, occupies only about a dozen frames. A leaf-hidden fruit is invisible in a median 78% of the frames in which it still hangs on the tree, two-thirds of the hidden fruits are ripe, and in FruitCount the basket is never in the front view. Neither task can be solved from the current frame: the policy must remember where hidden ripe fruits are and how many fruits it has picked.

6.1. Picking events from the tool-to-fruit distance (RQ2, RQ3)

Detection is exact over a physically determined plateau. Events detected from the ground-truth distance have precision, recall and F1 of exactly 1.000 for every radius between 1 and 4 cm, on the training set (where r was selected) and the validation set, with the correct pick count in every validation episode (Fig. 3a; Supplementary Table S2). At 0.5 cm the gripper does not always come that close (recall 0.628); beyond half the inter-fruit spacing, approaches to neighbouring fruits merge into spurious events (count exact 0.49 at 10 cm). The contact keyframe precedes detachment by a median of 4 frames and the release keyframe coincides with it. On the 200 perturbed DART-H demonstrations, precision is 0.998, recall 1.000 and the count is exact in 99% of episodes.

Robustness yields an accuracy specification for perception. Figure 3b–d and Supplementary Table S3 report detection on corrupted validation distances (5 seeds per level). Independent dropouts are nearly harmless (F1 1.000 with 30% of frames missing, since unknown frames inherit the last state), and a bias of −1 to +2 cm, as from a miscalibrated depth camera,

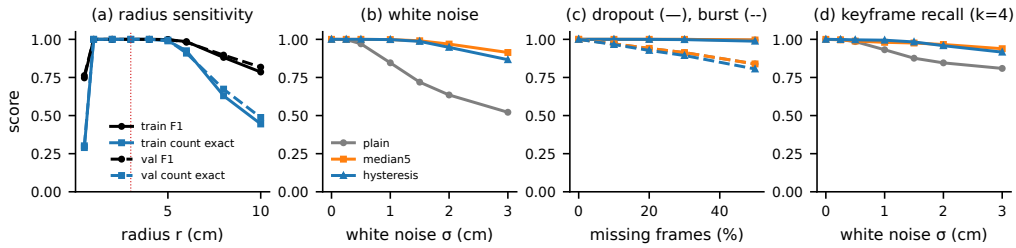


Figure 3: Event detection. (a) F1 and count exact rate versus radius r (training and validation). (b) F1 versus white-noise level for the plain, median5 and hysteresis detectors. (c) F1 versus the fraction of missing frames, independent (solid) and contiguous (dashed). (d) Recall of K4 memory keyframes (±3 frames) versus white-noise level.

leaves the count exact. White noise is the main threat: at 1 cm the plain detector's F1 falls to 0.846 and the count is exact in only 37% of episodes, because the distance chatters around the threshold. Both filters remove this (F1 0.999, count exact 99%); hysteresis does so without delay, keeping 0.995 of the K4 keyframes against 0.980 for the median filter, and is more robust to correlated noise (F1 0.953 versus 0.941 at 2 cm), so it is the default deployment detector. Long contiguous dropouts covering a whole approach remain the weak case (F1 0.894–0.913). With hysteresis, distance errors of up to 1.5 cm white noise (F1 0.987) or 2 cm correlated noise (F1 0.953) keep the event F1 above 0.95: this is the accuracy specification for any perception module that feeds PEKM.

6.2. What a fixed-budget memory retains (RQ4)

Table 4 compares memories at the 16-frame budget over 5318 decision points of the validation demonstrations (full table with all variants in Supplementary Table S4; an example of the selected frames in Supplementary Fig. S1).

Uniform memory loses picks systematically. It covers only 57.0% of the completed picks, only 43.1% of decisions can count the picks from memory, and even the most recent pick is covered in only 67.5% of decisions. Keeping one frame every 30 frames while a pick lasts about a dozen, it retains picks by chance, like random memory (43.2%); a recent-frame window is worse still (8.0%).

Forgetting grows with operating time, as the coverage analysis predicts. Over the 200 validation episodes (7884 decision–pick pairs), uniform memory

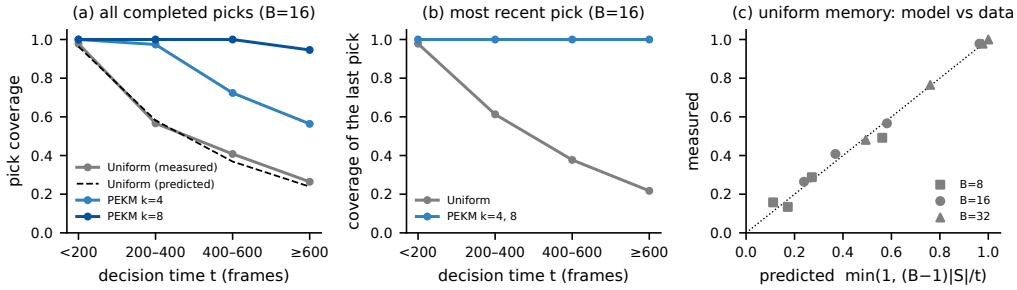


Figure 4: Time-indexed memory forgets more the longer the robot works. (a) Coverage of all completed picks versus decision time at $B = 16$: uniform memory (measured and predicted by Eq. (5)) and PEKM. (b) Coverage of the most recent pick. (c) Predicted versus measured coverage of uniform memory for every budget and decision-time bin.

at $B = 16$ covers 97.8% of completed picks in the first 200 frames, then 56.6%, 40.8% and only 26.4% beyond 600 frames, and the most recent pick falls from 97.8% to 21.7% (Fig. 4a,b). Equation (5) predicts the measured coverage to within 0.012 at every budget (predicted / measured: 0.282 / 0.270 for $B = 8$, 0.562 / 0.570 for $B = 16$, 0.882 / 0.883 for $B = 32$) and tracks its decay over time (Fig. 4c): the failure is structural and cannot be tuned away. PEKM removes the dependence on time: $k = 4$ and $k = 8$ cover the most recent pick at every decision time, and $k = 8$ covers at least 94.6% of all completed picks even beyond 600 frames.

PEKM retains recent picks by construction, without losing temporal context. With $k = 4$ the last two picks are always covered and overall coverage rises to 87.6%; with $k = 8$ coverage reaches 99.6% and count recoverability 99.2% (from 0.389 to 0.985 in FruitCount, from 0.485 to 1.000 in FruitOccludedRipe). The median gap between memory frames grows only from 14 to 17 frames (Fig. 5c). The soft variant concentrates six times more frames near fruits than U and covers older picks evenly, but at the cost of the most recent pick (0.814) and of temporal context (median gap 63 frames), which is why hard reservation is the main method. *Budget efficiency* follows: K4 with 8 frames (85.9%) exceeds uniform memory with 16 frames and approaches it with 32 (88.3%), and K8 with 16 frames (99.6%) exceeds uniform memory with 32 (Fig. 5a), i.e. the same event information with a memory two to four times shorter.

Table 4: Information retained by 16-frame memories (validation, both tasks; 5318 decisions, 7841 decision–pick pairs). Last / 2nd / 3rd: coverage of the most recent, second and third most recent pick. Near: fraction of memory frames with $d_t \leq r$. Gap: median gap between memory frames (frames).

Memory	Event cov.	Count rec.	Last	2nd	3rd	Near	Gap
Uniform (U, K0)	0.570	0.431	0.675	0.472	0.416	0.042	14
Recent	0.080	0.046	0.145	0.000	0.000	0.049	209
Random	0.432	0.329	0.533	0.351	0.231	0.040	41
PEKM $k = 2$	0.729	0.579	1.000	0.430	0.336	0.085	17
PEKM $k = 4$	**0.876**	**0.814**	**1.000**	**1.000**	0.136	0.110	17
PEKM $k = 8$	**0.996**	**0.992**	**1.000**	**1.000**	**1.000**	0.123	17
PEKM-S $\alpha = 1$	0.897	0.770	0.814	1.000	1.000	0.265	63

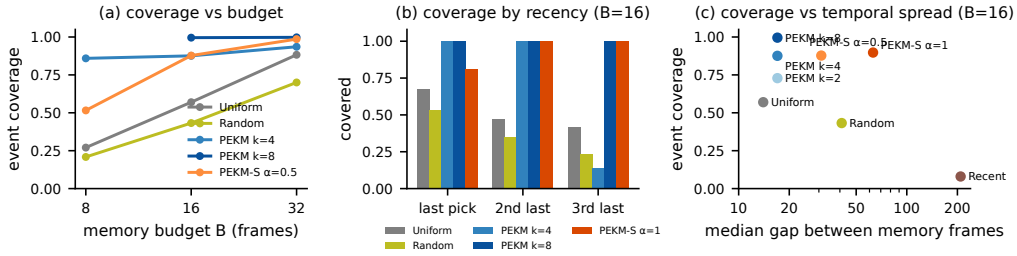


Figure 5: Memory coverage. (a) Event coverage versus budget B. (b) Coverage of the last three picks at $B = 16$. (c) Event coverage versus median gap between memory frames: PEKM raises coverage without sacrificing temporal resolution.

6.3. Fruit segmentation with SAM (RQ5)

Table 5 and Fig. 6 compare fruit segmentation on 8644 rendered validation images (13,424 front-camera and 4877 wrist-camera instances).

Prompted SAM gives the most accurate masks. SAM-FDM reaches an instance F1 of 0.991 (front) and 0.983–0.984 (wrist), with tighter masks than colour thresholding (IoU 0.974–0.978 versus 0.945), which matters because fruit centres are estimated from the mask, and recovers more partially hidden fruits (recall 0.967–0.976 versus 0.952 at 25–75% visibility, 0.633–0.688 versus 0.615 below 25%). Ripeness is correct for every matched instance with all methods.

Prompts are essential, and a small backbone suffices. Without prompts, SAM-AUTO is precise (0.994–0.996) but misses partially hidden fruits (recall 0.57–0.62 at 25–75% visibility, 0.21–0.26 below 25%) and is four to seven times slower (395–626 ms per image). The three SAM-FDM backbones have

Table 5: Fruit instance segmentation and ripeness on rendered validation images (8644 images; 13,424 front and 4877 wrist instances). F1 at IoU $\geq$ 0.5. IoU: mean over matched front-camera instances. R (vis.): front-camera recall by visible fraction (418 instances at 25–75%, 109 below 25%). Latency: median per front image on one L20 GPU. SAM-AUTO was evaluated on every 5th frame.

Method	Backbone	F1 front	F1 wrist	IoU	R (25–75%)	R (<25%)	Ripeness	ms/img
HSV thresholds	–	0.990	0.980	0.945	0.952	0.615	1.000	1.4
SAM-AUTO	ViT-B	0.937	0.983	0.979	0.573	0.263	1.000	395
SAM-AUTO	ViT-H	0.941	0.981	0.970	0.618	0.211	1.000	626
SAM-FDM	ViT-B	**0.991**	**0.984**	0.974	0.967	0.633	1.000	94
SAM-FDM	ViT-L	**0.991**	**0.984**	**0.978**	**0.976**	**0.688**	1.000	195
SAM-FDM	ViT-H	**0.991**	0.983	0.975	0.971	0.651	1.000	309
SAM-GTBOX (upper bound)	ViT-H	0.999	0.999	0.980	1.000	0.899	1.000	–

the same F1 (0.991) and differ by at most 0.004 in IoU, while latency grows from 94 ms (ViT-B) to 309 ms (ViT-H); since SAM runs once per episode, ViT-B is the practical choice for embedded platforms. With ground-truth boxes, SAM reaches F1 0.999 and a recall of 0.899 below 25% visibility, so the remaining gap lies in prompting barely visible fruits, not in SAM's masks.

Illumination locates the component that must adapt. We re-rendered every third frame under overcast or dusk, direct sun and warm low sun with the colour thresholds kept as calibrated (Fig. 6c; Supplementary Table S5). Given correct box prompts, SAM's F1 stays at 0.999 and its IoU at 0.979 under all three conditions, whereas fixed colour thresholds do not transfer (F1 below 0.1) and SAM-FDM inherits this through its prompts. The component to adapt for field use is therefore the prompt generator, which can be replaced by an illumination-normalized colour model or a learned detector without changing SAM, the anchors or the policy. [TBD: Optional: colour-normalized or learned prompt source as an additional row.]

6.4. From RGB-D images to event memory (RQ6)

Table 6 evaluates the complete chain on the 199 validation episodes using only the recorded front RGB-D frames, forward kinematics and the gripper state, with no simulator ground truth.

Anchors, distances and events. Frame-0 anchors recover 99.8% of fruits with 99.5% precision, a median localization error of 0.48 cm (90th percentile 0.58 cm), below a fifth of the fruit radius, and error-free ripeness; all 138 ripe fruits that the gust later hides have an anchor (Supplementary Table S6). The perceived distance has a median near-range error of 0.34 cm (90th percentile 0.48 cm), well inside the accuracy specification above. Events de-

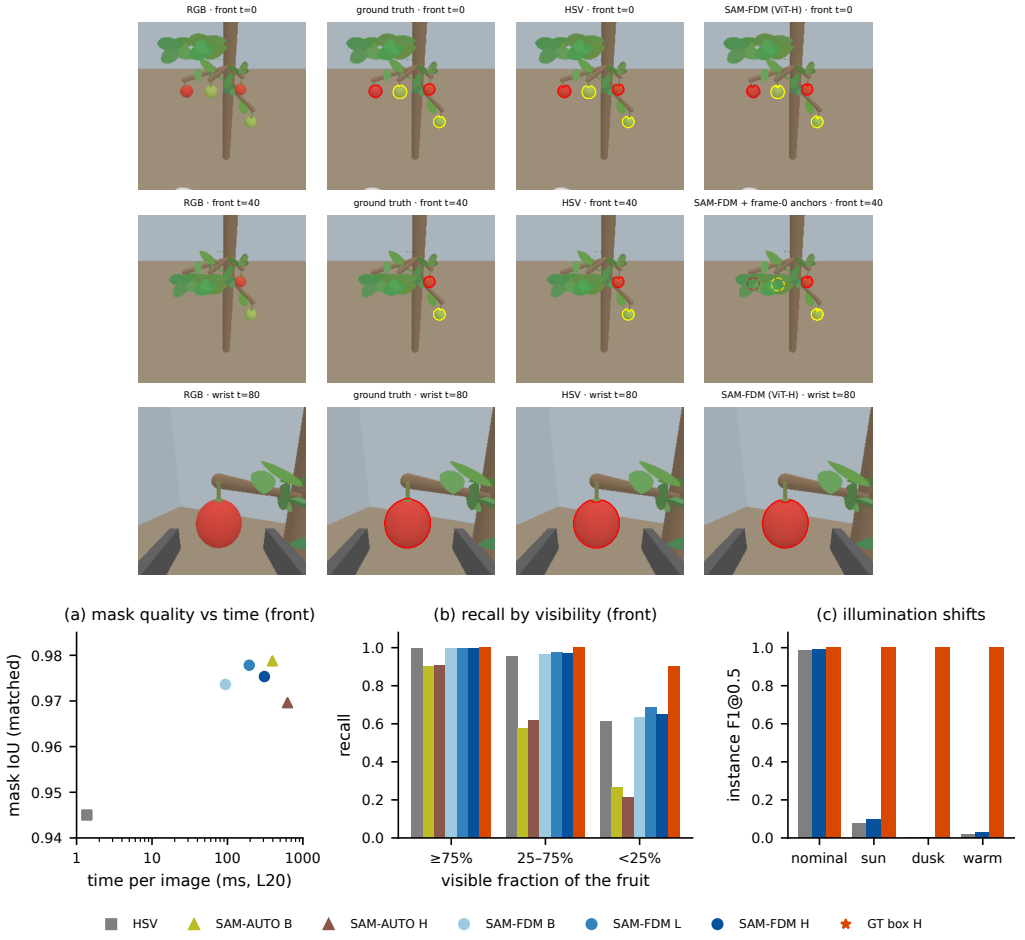


Figure 6: Fruit segmentation. Top: a FruitOccludedRipe validation episode before the gust (front, $t = 0$), after it (front, $t = 40$) and at contact (wrist, $t = 80$); columns show the image, ground truth, HSV thresholds and SAM-FDM (ViT-H), with red contours for ripe and yellow for unripe fruits. At $t = 40$, dashed circles are the frame-0 anchors, which keep the two leaf-hidden fruits in memory. Bottom: (a) mask IoU versus latency; (b) front-camera recall by visible fraction; (c) F1 under illumination shifts.

tected from $\hat{d}_t$ match the ground truth exactly (F1 1.000, count exact in every episode, release error 0 frames), and the PEKM memories recover every ground-truth keyframe (Jaccard 0.948 for $k = 4$, 0.941 for $k = 8$, the differences being contact frames shifted by one or two frames). Consumer-grade depth noise leaves all results unchanged.

Table 6: Events and memory from perceived distances (199 validation episodes; plain detector, hysteresis in parentheses). Dist. err.: median / 90th-percentile error of $\hat{d}_t$ where $d_t \leq 10$ cm. Rel. err.: median release-keyframe error (frames). Jaccard and KF recall: agreement of the K4 memory with the GT memory. For K8, Jaccard is 0.941 / 0.941 / 0.660 / 0.730 / 0.363 and KF recall 1.000 / 1.000 / 0.847 / 0.882 / 0.782 for the rows below GT.

Distance source	Dist. err. (cm)	P	R	F1	Count	Rel. err.	Jaccard	KF recall
GT (upper bound)	0 / 0	1.000	1.000	1.000	1.00	0	1.000	1.000
SAM-FDM (ours)	**0.34 / 0.48**	**1.000**	**1.000**	**1.000**	**1.00**	**0**	**0.948**	**1.000**
SAM-FDM + depth noise	0.31 / 0.46	1.000	1.000	1.000	1.00	0	0.948	1.000
Depth-only	0.39 / 10.76	1.000	1.000	1.000	1.00	16	0.685	0.845
Post-gust anchors	0.39 / –	1.000	0.743	0.853	0.50	0	0.741	0.880
Per-frame	0.39 / 12.79	0.821 (0.942)	1.000	0.902 (0.970)	0.62 (0.85)	52	0.421	0.755

Spatial memory before occlusion is necessary. With anchors built after the gust (frame 29), anchor recall drops to 0.783 (0.568 in FruitOccludedRipe), none of the 138 leaf-hidden ripe fruits receives an anchor, event recall in FruitOccludedRipe falls to 0.450, and *no* FruitOccludedRipe episode has the correct pick count (0.00, against 1.00 with frame-0 anchors); FruitCount, without foliage occlusion, is unaffected. Remembering where a fruit is, perceived while visible, is a precondition for remembering that it was picked.

Grasp evidence closes the visual blind spot, and anchors beat per-frame detection. With depth vacancy alone, the arm and the carried fruit occlude the anchor during carrying, so the release keyframe is late by a median of 16 frames and the memory Jaccard drops to 0.685; the nearly free gripper-holding cue removes the delay. Without anchors, the carried fruit is still detected as the nearest fruit: the release error grows to 52 frames and spurious events appear (precision 0.821, count exact in 62% of episodes; 0.759 and 46% in FruitCount), and even with hysteresis (precision 0.942) the memory Jaccard remains 0.421 (Fig. 7).

6.5. Closed-loop harvesting, ablations and counterfactual controls (RQ7)

Table 7 reports closed-loop results on validation episodes 50–99 (150 trials per task and condition). [TBD: Main results after the sealed evaluation: success of K0/K4/K8 per task with paired differences, 95% CIs and Holm-adjusted p-values; process metrics; stratification by quota N (2 / 3 / 4–5); validity gates.] Table 8 lists the test-time ablations of the K4 policy under the same paired protocol. [TBD: Ablation results: keyframe cap, radius, hard vs soft selection, signal source, counterfactual controls EVU/EVR/EVA, keyframe composition, training data.]

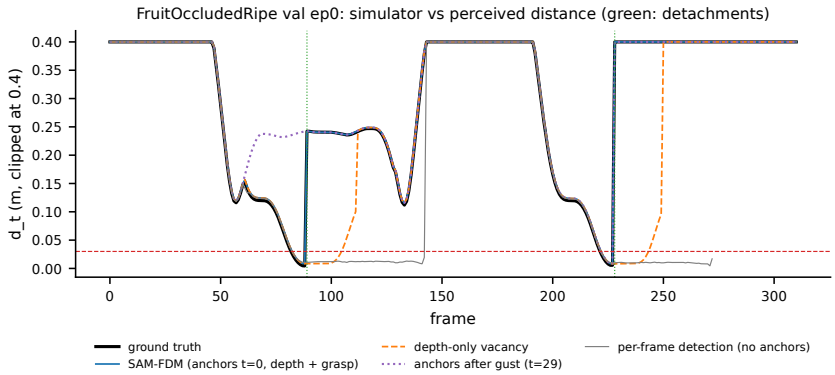


Figure 7: Perceived versus ground-truth distance in one FruitOccludedRipe episode (vertical lines: detachments). SAM-FDM follows the ground truth; depth-only vacancy confirms detachment late, post-gust anchors miss the hidden ripe fruit, and per-frame detection tracks the carried fruit.

Table 7: Closed-loop harvesting on validation episodes 50–99 (3 seeds; 150 trials per cell). Paired differences with 95% bootstrap CIs; Holm-adjusted p-values over the primary family.

Task	Memory	Success	Completion	Over-/wrong-pick	Drop	Steps	Δ vs K0 [95% CI]
FruitCount	K0 (uniform)	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	–
	K4 (PEKM)	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
	K8 (PEKM)	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
FruitOccludedRipe	K0 (uniform)	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	–
	K4 (PEKM)	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
	K8 (PEKM)	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]

Table 8: Ablations (test-time variants of the K4 policy unless noted; paired differences to the default configuration).

Factor	Setting	FruitCount Δ	FruitOccludedRipe Δ
Keyframe cap k (trained)	0 / 4 / 8	[TBD:]	[TBD:]
Radius r	1 / 3 / 5 cm	[TBD:]	[TBD:]
Selection	hard (PEKM) / soft (PEKM-S, $\alpha = 0.5$)	[TBD:]	[TBD:]
Signal source	GT / SAM-FDM / GT + 1 cm noise	[TBD:]	[TBD:]
Counterfactual	EV / EVU / EVR / EVA	[TBD:]	[TBD:]
Keyframes	contact / release / both	[TBD:]	[TBD:]
Training data	clean (C0) / clean + DART-H (K0)	[TBD:]	[TBD:]

6.6. Supporting analyses: closed-loop drift, memory content and cost (RQ8)

Open-loop accuracy and closed-loop drift. At the demonstrated states, a uniform-memory policy trained on clean demonstrations has median TCP errors of 1.7–2.4 cm at the end of a 16-step chunk and 0.7–1.1 cm near the fruit, an order of magnitude below the motion within a chunk, and agrees

with the demonstrated gripper direction in 98–99.5% of decisions, with nearly identical training and validation errors (Supplementary Table S7). In closed loop, small deviations accumulate into states that clean demonstrations do not cover; DART-H perturbations (median 2.5 cm) match this scale. [TBD: Closed-loop C0 vs K0 comparison.]

Memory content matters for VLA policies. On a general manipulation-memory benchmark ([TBD: cite RoboMME]; sealed protocols, Supplementary Table S8), removing the history memory of the base policy collapses success on a pattern-recall task from 66.7% to 1.7% (−65.0 pp, 95% CI [−81.7, −46.7]), halving the budget from 32 to 16 frames reduces it from 64.4% to 26.7%, and replacing a single memory slot changes the action loss with a slot-ranking reliability of 0.97–0.99 across four tasks: within a fixed budget, the choice of frames matters.

Run-time cost. At $t = 1000$ frames, one PEKM selection takes 117 μs on a CPU core (39 μs for uniform sampling), negligible against one VLA inference ([TBD: VLA latency, ms]), and the memory footprint and inference cost of the policy are those of the uniform baseline. SAM-FDM runs SAM once per episode (94 / 195 / 309 ms for ViT-B / L / H on one L20 GPU), and its per-frame work, projecting a few anchors and reading depth and gripper state, takes well under a millisecond.

7. Discussion

From time-indexed to event-indexed memory. The memory of an embodied agricultural agent should be organized around what happens, not around when it happens. The coverage analysis (Fig. 4) shows that the failure of uniform memory is structural: for any fixed budget, the fraction of events it retains vanishes as operation lengthens, precisely the regime of a robot working continuously along a row. Event indexing removes this dependence, because picks are short but physically unambiguous and two slots per remembered pick suffice. The approach–act–leave structure is not specific to fruit picking: pruning, thinning, pollination and spot spraying all consist of discrete interactions with plant organs, for which the tool-to-organ distance is an equally natural event signal, so event-centric memory should transfer with only a redefinition of the target set $\mathcal{P}_t$.

Designing perception for memory. First, *perception must meet a specification, not a maximum*: event memory tolerates about 1.5–2 cm of distance

noise, dropouts and bias, and SAM-FDM is an order of magnitude more accurate, leaving room for cheaper detectors or sensors behind the same scalar interface. Second, *memory starts with perception*: leaf-hidden ripe fruits are recoverable only if they were anchored while visible, which argues for a deliberate observation phase, or anchoring fruits whenever they become visible, and for redundant evidence of picking. The gripper-holding cue removed a 16-frame delay at essentially zero cost, so proprioceptive and tactile signals should be treated as inputs to event detection, not only as control feedback.

Practical implications and agricultural foundation models. For a gripper that must enclose a 5-cm fruit, a few centimetres of closed-loop drift separate a grasp from a miss, and recovery data generated without an expert and restricted to safe phases addresses the data rather than the model. [TBD: Discussion of the closed-loop DART-H and PEKM results.] Because the selection is a pure function of a logged signal, every memory the policy saw can be reproduced and audited. More broadly, as foundation models enter agricultural robotics, what they remember will matter as much as what they perceive. HarvestMem shows that a running account of treated plants, hidden fruits and open quotas can be supplied without changing the model's architecture or adding parameters: the agricultural knowledge is encoded in *what counts as an event*, which domain experts can specify and audit.

Limitations and future work. The study is conducted in simulation, with rigid spherical fruits, a fixed stem-detachment rule, simplified branch mechanics and wind only during observation. Colour prompts do not transfer to strongly different illumination, and green fruits on green foliage, natural light and clustered fruits will require learned or illumination-normalized prompt generators and an event definition for clusters. One tree type and one arm were used in closed loop, and real-orchard transfer on an RM75-B with an RGB-D camera remains to be tested. [TBD: Limitations revealed by the closed-loop results.] Natural extensions are giving the detected pick count to the policy as a text prompt (e.g. "picked 2 of 3"), a memory manager learned from counterfactual action-loss signals, anchor-guided leaf pushing, and sim-to-real transfer with randomized illumination and foliage.

8. Conclusions

This paper showed that the time-indexed memory of current vision–language–action policies is structurally unsuited to selective harvesting, and

proposed HarvestMem, which couples a spatial memory of 3D fruit anchors, acquired by promptable segmentation while fruits are visible, with an episodic memory that reserves the contact and detachment keyframes of picks in the policy's fixed memory budget. The main findings are:

1. Uniform memory covers completed picks as $c_U(t) \approx \min(1, (B-1)|S|/t)$ (to within 0.012); at 16 frames, coverage of the most recent pick falls from 97.8% to 21.7% after 600 frames.
2. Picking events are detected exactly from the end-effector-to-fruit distance over a 1–4 cm range of the radius, on clean and perturbed demonstrations alike.
3. Picking-Event Keyframe Memory retains 99.6% of completed picks against 57.0% for uniform memory, keeps the most recent pick at every decision time, and with 8 frames outperforms uniform memory with 16, at no additional cost.
4. From RGB-D images and proprioception, the SAM-based fruit distance module localizes fruits to 0.48 cm, recovers every leaf-hidden ripe fruit and reproduces the ground-truth events exactly; without anchors acquired before occlusion, no occluded-task episode is counted correctly.
5. Event memory tolerates 1.5–2 cm distance errors, dropouts and bias; prompted SAM gives the most accurate fruit masks, and its prompt generator is the component to adapt to field illumination.
6. [TBD: One sentence on closed-loop success with PEKM and DART-H (E7–E9).]

By moving agricultural knowledge into the definition of events rather than into network parameters, HarvestMem offers a principled, auditable way to adapt robot foundation models to the long, partially observable operations of agriculture.

CRediT authorship contribution statement

[TBD: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Funding acquisition.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work

reported in this paper.

Data availability

Supplementary tables, figures and definitions accompany this article. The HarvestMem-Sim generator, demonstrations, sealed evaluation protocols (with SHA-256 digests), analysis scripts and raw per-episode results will be released at [TBD: repository URL] upon publication.

Acknowledgements

[TBD: Funding sources and grant numbers.]

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